

fp-tools: A Reproducible Platform for ATAC-seq Footprinting and Regulatory Motif Analysis

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Abstract

Transcription-factor (TF) footprinting from ATAC-seq provides a base-resolution view of regulatory factor occupancy, yet complete analyses often require multiple tools with different interfaces, signal conventions, and reproducibility practices. We present *fp-tools*, a standalone Python package that integrates Tn5 bias correction, footprint calling, motif matching, differential footprint analysis, and aggregate visualization within a consistent command-line workflow. The same framework supports de novo motif-discovery preparation, pseudobulk aggregation from single-cell ATAC fragments, and replicate-aware reporting for condition comparisons. Using public bulk ATAC-seq replicates and 10x Genomics PBMC Multiome data, we show that *fp-tools* recovers biologically interpretable motif-centered aggregate patterns while preserving a reproducible analysis trail through versioned inputs, source tables, and regression-tested command outputs. *fp-tools* provides an integrated and auditable platform for classical ATAC-seq footprinting workflows and their motif-centered extensions.

Keywords: ATAC-seq; transcription factor footprinting; chromatin accessibility; motif analysis; pseudobulk ATAC-seq; replicate-aware reporting; reproducibility; command-line bioinformatics

1. Introduction

Chromatin accessibility profiling by the assay for transposase-accessible chromatin using sequencing (ATAC-seq) has become a standard readout of the active regulatory landscape of a cell [1]. Within accessible regions, sequence-specific transcription factors (TFs) protect the DNA they occupy from transposition, leaving characteristic depletions—“footprints”—flanked by hyper-accessible edges. Detecting and interpreting these footprints supports inference of TF occupancy, differential regulation between conditions, and the functional consequences of non-coding variation.

A mature but *fragmented* ecosystem of methods addresses pieces of this problem. Classical footprinting and differential-binding frameworks such as TOBIAS [2] and HINT-ATAC [3] correct enzymatic sequence bias and score footprints at a single TF-relevant scale. Multiscale and nucleosome-aware approaches, exemplified by the PRINT/scPrinter family [4], model accessibility structure across a range of footprint widths. Supervised TF-binding-site (TFBS) predictors such as maxATAC [5] and sequence-to-profile models such as ChromBPNet [6] learn occupancy from labeled data or raw sequence, and downstream interpretation tools and variant scorers translate models into motif and allele-level effects. Each tool typically ships its own interface, input conventions, and assumptions, so

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assembling a coherent, reproducible pipeline—and *benchmarking* alternatives on common ground—remains laborious and error-prone.

Beyond this tooling friction, footprint interpretation has practical limitations that make reproducibility and careful signal handling important. Classical footprinting depends on base-resolution cut-site depletion and can be confounded by enzyme sequence bias, local accessibility, motif strength, and factor-specific binding kinetics [7]. Deep sequence-to-profile models and multiscale methods can improve sensitivity for some questions, but they often require substantial compute and specialized training workflows. For many applied ATAC-seq projects, the immediate need is therefore a transparent, repeatable workflow that keeps the classical analysis reliable while exposing selected extensions without forcing users into a different software ecosystem.

Four gaps motivate this work. First, **signal clarity**: users need to know which signal representation is appropriate for footprint calling and how optional normalization changes interpretation. Second, **interface fragmentation**: moving from bias correction to footprint calling to differential footprint analysis to visualization commonly spans several packages with incompatible I/O. Third, **reproducibility and testing**: many footprinting workflows lack deterministic regression tests, pinned environments, and machine-checkable output contracts, making it hard to know whether a change altered results. Fourth, **extensibility without regression**: adding new scientific behavior, such as multiscale scoring or pseudobulk aggregation, should not destabilize the classical workflow.

We address these gaps with *fp-tools*, a reproducible ATAC-seq footprinting platform that connects bias correction, footprint calling, motif matching, differential footprint analysis, and aggregate visualization through a single command-line workflow. The package also supports de novo motif-discovery preparation, pseudobulk ATAC aggregation, and replicate-aware reporting, allowing these analyses to be incorporated without changing the core signal-processing path. This paper describes the implemented methods, public-data validation examples, and reproducibility framework that tie these workflows together.

2. Materials and Methods

2.1. Package Architecture and Core Workflows

fp-tools is distributed as a pip-installable package (*fp-tools-bio*) with vendored Cython internals for performance-critical signal operations. The core workflow has four method layers: Tn5 sequence-bias correction, footprint scoring over corrected signal, motif-anchored differential TF-binding analysis, and aggregate visualization around TFBS or region sets. These layers share common region, signal, motif, and sample-label conventions so intermediate files can be passed through a complete analysis without format-specific glue code. YAML configuration files provide a reproducible way to save and rerun the same analysis without changing the underlying methods.

For classical footprint calling, let x_i denote corrected Tn5 cut-site signal around candidate position p . The default footprint score is a center-versus-flank contrast,

$$F(p) = \frac{1}{|L| + |R|} \left(\sum_{i \in L} x_i + \sum_{i \in R} x_i \right) - \frac{1}{|C|} \sum_{i \in C} x_i,$$

where C is the protected center and L and R are flanking windows. Larger values therefore correspond to the expected footprint pattern: fewer cuts at the putative bound site and more cuts in nearby accessible flanks.

2.2. Optional Analysis Modules

Several analysis modules extend the core footprinting workflow:

- *De novo motif-discovery preparation* exports candidate-centered sequences, records reproducible MEME/DREME [8] and Tomtom [9] settings, and summarizes discovered motifs and matches to known motif databases. This can be run as a standalone discovery step or as a supplement to known-motif analyses when a database, such as a JASPAR release, may not contain all active motifs in the sample.
- *Pseudobulk aggregation* groups single-cell ATAC fragments by annotation columns into bulk-like fragment files, manifests, optional indexed outputs, and optional normalized cut-site tracks for downstream footprinting and aggregate plots.
- *Replicate-aware differential footprint analysis* treats repeated samples from the same condition as biological replicates, reports condition means and replicate variation, and produces diagnostic tables and figures for differential TF-binding comparisons (Section 2.4).

Conceptually, the optional modules answer focused questions around the classical core. De novo motif preparation asks which sequence patterns recur among exported candidate sites; replicate-aware reporting asks whether repeated-condition inputs can be summarized as condition means and replicate variation; and pseudobulk aggregation asks whether single-cell fragments can be grouped into interpretable group-level profiles that run through the same footprinting machinery. In this work we validate that path with a public 10x Genomics PBMC Multiome dataset processed by Cell Ranger ARC 2.0.0 [10]. Official 10x gene-expression graph clusters are collapsed into broad PBMC labels using canonical marker-gene enrichments (B cells, CD4 T cells, NK/cytotoxic T cells, CD14 monocytes, FCGR3A monocytes, dendritic cells, and mixed myeloid cells), then ATAC fragments are split by label and converted to CPM-normalized cut-site bigWigs. For aggregate validation, exact motif-centered site sets are generated by scanning JASPAR2026 CORE vertebrate motifs inside the 10x ATAC peak set. A predeclared immune-regulator panel is screened by expected-lineage center protection, and the displayed panels are selected from that source table rather than from peak centers.

2.3. Public ATAC Processing and Reproducibility Provenance

The bulk replicate demonstration uses true biological replicates from the Buenrostro et al. ATAC-seq study [1]: GM12878/B-cell samples SRR891268 and SRR891269, and day-1 CD4 T-cell samples SRR891275 and SRR891276. Paired-end FASTQ files were adapter-trimmed, aligned to hg38, filtered for proper high-quality pairs, duplicate-marked, blacklist-filtered, peak-called per replicate, merged into a common peak universe, bias-corrected, footprint-called, and analyzed by differential footprint comparison with and without sample-level quantile normalization. The analysis used JASPAR2026 CORE vertebrates non-redundant motifs for known-motif scanning. Exact command-line parameters are recorded in a source TSV in the manuscript tables directory; the main settings are summarized in Table 1.

Table 1. Primary analysis parameters used for the public ATAC and pseudobulk workflows. The complete tab-separated source table is stored as `analysis_parameters.tsv`.

Step	Parameters
FASTQ preprocessing	Paired-end adapter detection with <code>fastp</code> ; HTML and JSON QC reports written per sample.
Alignment and BAM filtering	<code>bowtie2 -very-sensitive -X 2000</code> against hg38; retain proper pairs with <code>samtools view -f 2 -F 2828 -q 30</code> ; sort, fixmate -m, coordinate sort, markdup -r, index, flag-stat, and idxstats.
Blacklist filtering	Exclude mitochondrial and nonstandard contigs and subtract hg38 blacklist v2 regions.
Peak calling and union peaks	MACS3 paired-end peak calling with human genome size, duplicate retention, and q-value 0.01; replicate peaks are sorted and merged into one analysis peak set.
Bias correction and footprint calling	<code>atac-correct</code> with merged peaks and blacklist using default Tn5-bias settings; <code>call-footprints -score footprint</code> with default footprint/flank windows.
Differential footprint analysis	Four replicate footprint tracks analyzed with condition labels Bcell, Bcell, Tcell, Tcell; JASPAR2026 vertebrate non-redundant motifs; replicate report enabled; runs performed with no normalization and with sample-level quantile normalization.
Pseudobulk PBMC workflow	10x PBMC Multiome fragments grouped by broad immune labels; groups retained with at least 300 cells and 50,000 fragments; CPM-normalized cut-site bigWigs generated for aggregate profiles.

All software versions used for the manuscript analyses are recorded in a source TSV in the manuscript `tables` directory and summarized in Table 2. This table includes the FASTQ-processing tools, fp-tools runtime, Python scientific stack, motif database release, and external source-processing version for the 10x PBMC dataset.

Table 2. Software and resource versions used for manuscript analyses. The complete source table is stored as `software_versions.tsv`.

Tool or resource	Version	Role
fp-tools	0.1.7	Footprinting, motif analysis, pseudobulk, and plotting utilities
Python	3.12.3	Runtime for analyses and figure builders
fastp	1.3.3	FASTQ trimming and QC
bowtie2	2.5.5	hg38 read alignment
samtools/htslib	1.23.1/1.23.1	BAM filtering and QC
bedtools	2.31.1	Region sorting, merging, and blacklist subtraction
MACS3	3.0.4	ATAC-seq peak calling
NumPy/pandas/SciPy	2.4.6/2.3.3/1.17.1	Numerical and tabular analysis
matplotlib/pyBigWig	3.10.9/0.3.25	Figures and bigWig signal I/O
pysam/Biopython	0.23.3/1.87	BAM I/O and sequence utilities
JASPAR CORE vertebrates	2026 non-redundant	Known motif database
MEME	5.5.9	De novo motif discovery and discovered-motif matching
Suite/STREME/Tomtom		
Cell Ranger ARC	2.0.0	Source 10x PBMC Multiome processing
hg38 reference and blacklist	GRCh38/hg38; blacklist v2	Alignment, sequence extraction, and artifact filtering

2.4. Replicate-aware Differential Binding and Quantile Normalization

Replicate-aware analysis is integrated directly into the differential footprint comparison workflow. If users pass repeated condition names, for example `-cond-names Bcell Bcell Tcell Tcell`, fp-tools treats the corresponding signal files as biological replicates of the same condition while preserving the original differential-binding output columns. Let $x_{c,r,i}$ be the footprint score at site or background position i for replicate r of condition c . With `-normalization sample-quantile`, each replicate is normalized before condition averaging using the same multiplicative quantile curve used throughout fp-tools. For sample s , fp-tools computes empirical positive score quantiles $q_s(p)$ and the reference quantile $q_*(p) = S^{-1} \sum_s q_s(p)$ across the S input samples, then fits a smooth multiplicative factor $g_s(x)$ to the ratio $q_*(p)/q_s(p)$. Each score is transformed as

$$\tilde{x}_{s,i} = \max\{0, x_{s,i} g_s(x_{s,i})\}.$$

The default condition-quantile mode preserves the previous behavior by first averaging replicate background distributions within each condition, fitting one factor $g_c(x)$ per condition, and applying that condition-level factor to all replicates in the condition before final condition means and SDs are reported; none disables this step. The same shared

normalization helper and the same sample-first versus condition-first ordering are used by plot-aggregate, so aggregate visualizations and differential footprint comparisons are normalized consistently. Because this step removes distribution-level depth and background offsets, differential interpretation is based on normalized per-site contrasts and the differential-binding comparison statistic, not on global baseline height alone.

After normalization, condition-level scores are summarized as

$$\bar{x}_{c,i} = \frac{1}{n_c} \sum_{r=1}^{n_c} \tilde{x}_{c,r,i}, \quad s_{c,i} = \sqrt{\frac{1}{n_c - 1} \sum_{r=1}^{n_c} (\tilde{x}_{c,r,i} - \bar{x}_{c,i})^2}.$$

For a comparison between conditions c_1 and c_2 , fp-tools reports both an interpretable footprint-score difference and a stabilized log fold change,

$$\Delta_i = \bar{x}_{c_1,i} - \bar{x}_{c_2,i}, \quad L_i = \log_2 \frac{\bar{x}_{c_1,i} + \epsilon}{\bar{x}_{c_2,i} + \epsilon},$$

where ϵ is the pseudocount estimated from the background signal unless supplied explicitly. When both conditions have replicate support, approximate standard errors are reported as

$$SE(\Delta) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

$$SE(L) \approx \frac{1}{\ln 2} \sqrt{\frac{s_1^2}{n_1(\mu_1 + \epsilon)^2} + \frac{s_2^2}{n_2(\mu_2 + \epsilon)^2}}.$$

For each comparison, fp-tools reports the standard change and pvalue columns together with a Benjamini–Hochberg-adjusted qvalue_bh column and a significant_fdr05 flag. The existing highlighted column is retained as a report-selection and visualization flag, not as a formal multiple-testing-corrected significance label. Additional replicate-oriented columns include replicate counts, condition score SDs, mean Δ , mean L , and standard-error summaries.

2.5. Public Data, Benchmark Tiers, and Metrics

Benchmarks are organized as versioned TSV manifests recording source, benchmark tier, cell type, donor, TF, assay, accessions, assembly, output type, format, URL, checksum, status, local path, split, and notes. A discovery script queries the ENCODE REST API for released human GRCh38 ATAC-seq and matched TF ChIP-seq/CUT&RUN; a resumable downloader writes per-benchmark download reports with checksums and local paths. Raw and processed public data, and large benchmark outputs, are kept out of version control; scripts, manifests, schemas, small reference inputs, and figure code are tracked.

The benchmark is organized as a tiered design (Table 3): a small-reference regression tier; a bulk matched tier (ENCODE bulk ATAC vs. matched TF ChIP/CUT&RUN) for classical benchmarking; a depth-robustness tier over downsampled ATAC; a de novo motif-preparation tier; and a real-data pseudobulk demonstration tier. For binary TF-binding labels we compute AUROC and AUPRC globally and per TF-cell task, recall at 1%, 5%, and 10% FDR, precision at fixed recall, calibration curves with expected and maximum calibration error, Brier score for probability models, per-TF macro averages, Wilcoxon signed-rank tests for paired method comparisons, and stratified bootstrap confidence

intervals for key deltas. Expected calibration error partitions predictions into B equal-width probability bins and averages the gap between confidence and accuracy,

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{n} |\text{acc}(b) - \text{conf}(b)|,$$

with maximum calibration error defined analogously as the worst-bin gap. Metric, calibration, and bootstrap computations are implemented as standalone, unit-tested scripts and combined by a benchmark pipeline runner that also emits PDF/SVG/PNG figure panels.

Table 3. Tiered benchmark design. Large data and outputs are untracked; only manifests, scripts, and compact reference results are committed.

Tier	Purpose	Primary outputs
Small reference	Installation and regression checks on compact inputs	command success, output existence, stable summaries
Bulk matched	Main classical benchmark (ATAC vs. matched TF ChIP/CUT&RUN)	TFBS labels, predictions, AUROC/AUPRC, reports
Depth robustness	Shallow-depth sensitivity (50/25/10/5%)	performance-vs-depth and calibration curves
De novo motif prep	Candidate FASTA and motif-discovery validation	exported FASTA, command plans, motif summaries
Pseudobulk	Single-cell extension and real-data demonstration	10x PBMC cell-type pseudobulks, TF aggregate profiles

2.6. Reproducibility Engineering

The package ships a unit-test suite covering numerical helpers, CLI behavior, and deterministic golden CLI regressions for footprint calling, differential footprint analysis, and aggregate plotting, with an opt-in slow regression for bias correction. A continuous-integration workflow runs the suite on a clean checkout using compact reference inputs, builds source and wheel artifacts, and verifies the installed console scripts. The repository includes `environment.yml`, a Dockerfile, release metadata, and Make targets for test, build, and manuscript-smoke workflows; detailed reviewer instructions are kept in `docs/reproduce-paper.md`. Every benchmark result is expected to store random seeds, tool versions, command lines, and the resolved manifest. Figures are saved as vector PDF/SVG plus 300-dpi PNG previews, with source plotting tables retained for auditability.

3. Results

3.1. A Unified, Reproducible Footprinting Platform

`fp-tools` consolidates the four core ATAC footprinting workflows behind a single, consistent workflow with shared region/signal conventions, so a complete analysis—bias correction, footprint calling, differential footprint analysis, and aggregate visualization—runs without crossing package boundaries or reformatting intermediates. Optional YAML configuration records the same analysis in a machine-readable form for reruns and batch processing (Figure 1).

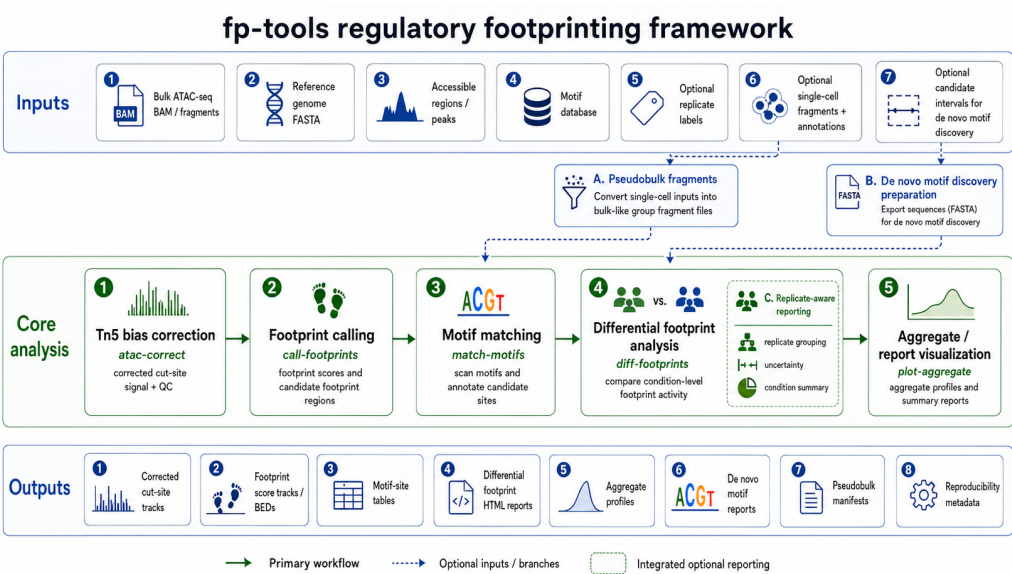
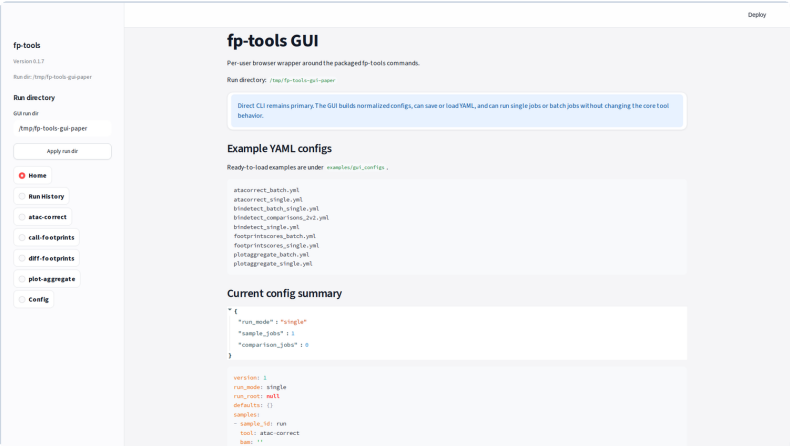


Figure 1. Generic fp-tools regulatory-footprinting framework. Bulk ATAC-seq inputs enter an integrated classical core for Tn5 bias correction, footprint calling, motif matching, differential footprint analysis, and aggregate visualization. Optional branches support single-cell pseudobulk fragment generation, de novo motif-discovery preparation from candidate intervals, and replicate-aware reporting within differential comparisons. Outputs include corrected cut-site tracks, footprint score tracks or candidate BEDs, motif-site tables, differential-footprint HTML reports, aggregate profiles, pseudobulk manifests, de novo motif reports, and reproducibility metadata.

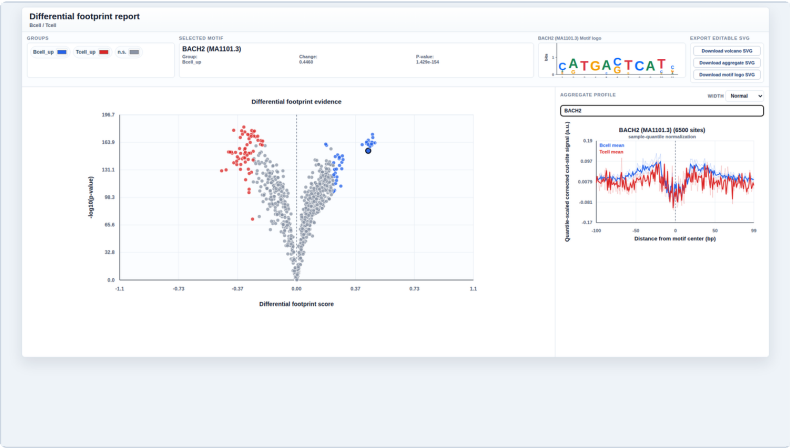
3.2. Interactive Interfaces and Report Review

In addition to direct command-line execution, fp-tools provides two usability layers for reviewing results. The optional GUI installed with fp-tools-bio[gui] builds the same normalized YAML configurations used by run-workflow, so browser-based setup remains compatible with the command line. Separately, the differential-footprint and aggregate-batch commands write standalone HTML reports with embedded compressed data, searchable motif controls, editable group colors, synchronized aggregate profiles, and SVG export for publication-ready editing (Figure 2).

A. Optional GUI wrapper



B. Interactive differential-footprint HTML



C. Interactive aggregate-batch HTML

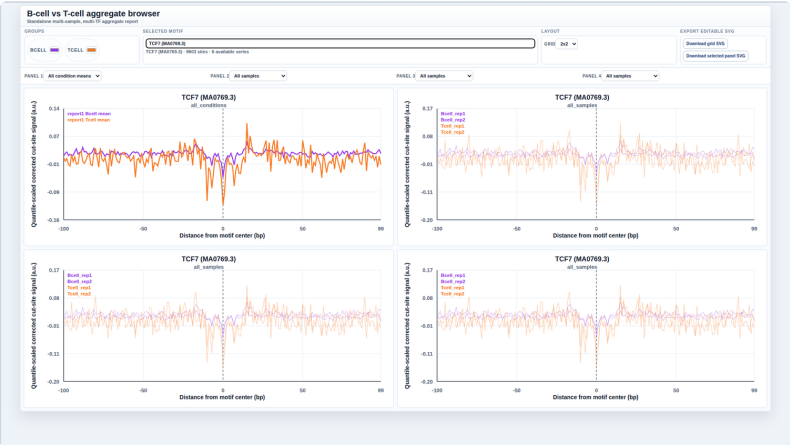


Figure 2. Usability-oriented fp-tools interfaces. (A) Optional browser GUI for configuring command-line workflows and saving runnable YAML configurations. (B) Standalone diff-footprints HTML report from the Buenrostro B-cell versus T-cell replicate comparison, combining differential evidence, selected motif metadata, motif logo display, and aggregate footprint profiles. (C) plot-aggregate-batch HTML report generated from the same embedded aggregate payload, allowing multi-sample and multi-TF aggregate review with searchable motif selection, group color controls, configurable panel layouts, and editable SVG export.

3.3. Regression-tested Workflow Outputs

The classical workflow is covered by deterministic regression tests that pin summary outputs for footprint calling, differential footprint analysis, and aggregate plotting. Public

entry-point checks and YAML dry-runs verify that the command surface remains stable. These tests provide a reproducible contract for core analyses while allowing optional modules to share the same input and output conventions.

3.4. De Novo Motif Discovery Complements Database-based Motif Analysis

The de novo motif-preparation path exports candidate-centered FASTA files, constructs reproducible MEME/DREME/STREME and Tomtom analysis records, and summarizes motif results into TSV/HTML reports with inline consensus logos. This keeps external motif discovery reproducible while relying on established motif-discovery algorithms. The workflow can be used alone to discover recurring patterns in candidate intervals, or after known-motif scanning to find additional motifs not represented in the selected database.

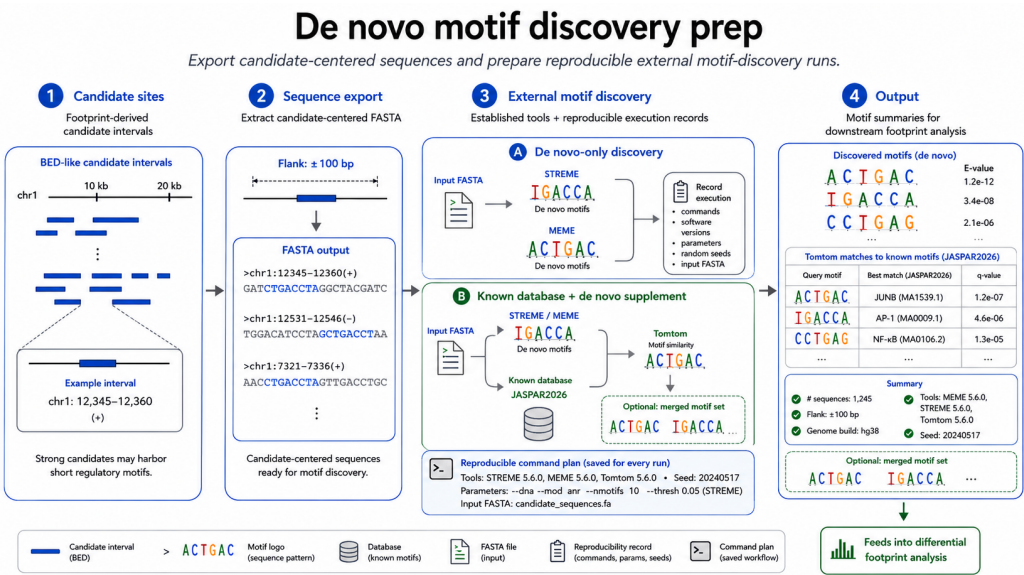


Figure 3. Generic de novo motif-discovery preparation workflow. Candidate footprint intervals are converted into candidate-centered FASTA sequences, external motif-discovery tools such as STREME or MEME are run with recorded parameters and software versions, and discovered motifs are summarized alone or matched back to known motif databases such as JASPAR2026. The same output can be used as a standalone de novo result or as a supplemental motif set for downstream differential footprint analysis.

We therefore evaluated whether candidate-derived motifs add information beyond a fixed database using the same 2-vs-2 Buenrostro B-cell versus T-cell ATAC-seq comparison used for replicate-aware reporting. Candidate footprints were called from bias-corrected cut-site tracks, condition-specific candidate FASTA files were exported with 75 bp flanks, and STREME 5.5.9 was run in both directions: B-cell candidates with T-cell candidates as control, and the reverse. The discovered motifs were then used in two ways. First, they were tested alone as a de novo-only motif set. Second, they were appended to known motif databases before running the same sample-quantile-normalized differential-footprint analysis.

The full JASPAR2026 vertebrate non-redundant set contained 1019 motifs, while the de novo procedure contributed 16 candidate-derived motifs. In de novo-only mode, several discovered motifs had footprint-like center depletion in the embedded aggregate profiles; these aggregate examples were selected by negative center-versus-flank scores rather than by database annotation. In supplement mode, the discovered motifs were appended to known motif sets before the same motif-aware differential footprint analysis. To create a sensitivity test, we also constructed a restricted JASPAR2026 set with 860 motifs and reran

the comparison with and without the 16 de novo motifs. The restricted database yielded 97 highlighted differential motif families; adding the de novo supplement increased this to 101 highlighted families, including 6 de novo-derived highlighted motifs. These analyses show the intended use case: when a chosen motif database is incomplete or conservative, candidate-derived motifs can be evaluated on their own and can also be carried forward into the same differential footprint report as database-supported motifs (Figure 4).

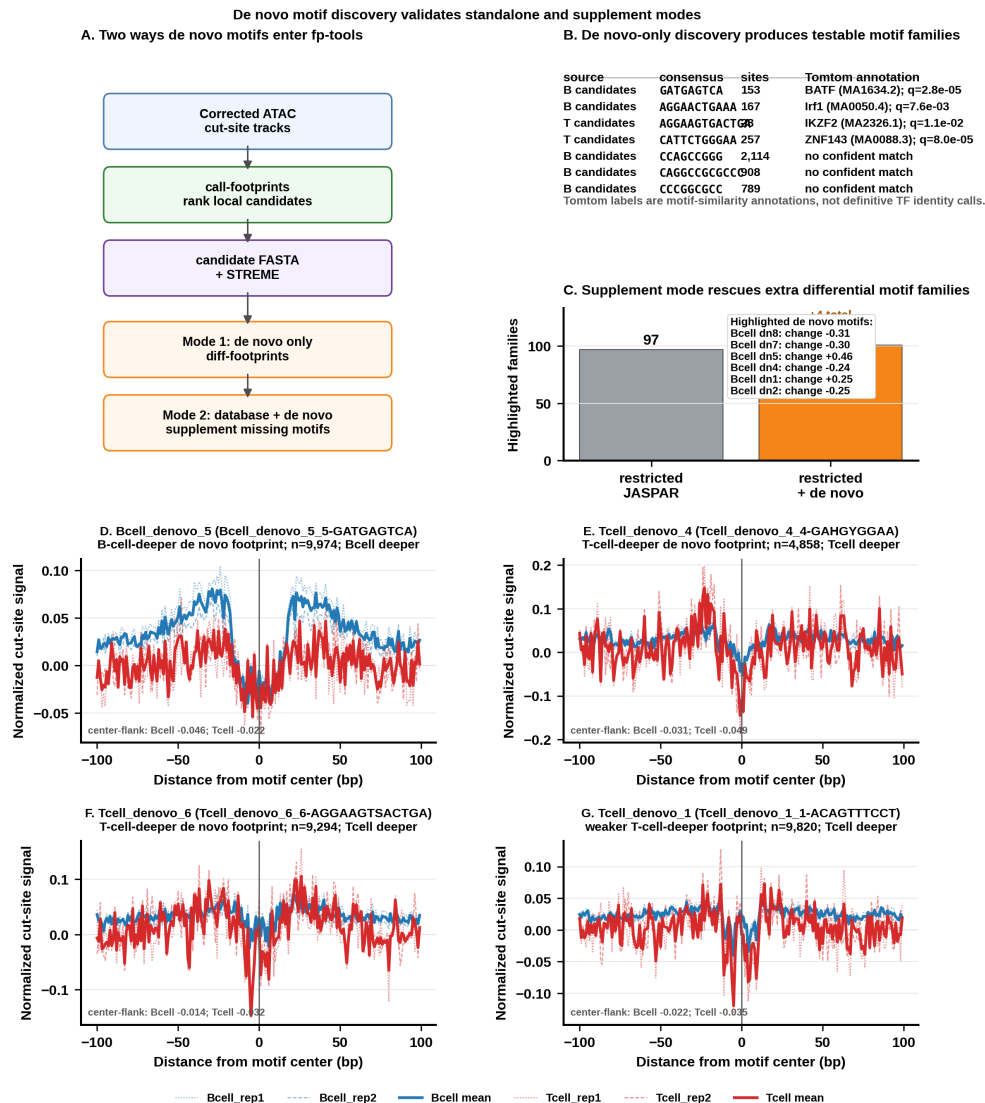


Figure 4. De novo motif discovery used as standalone analysis and as a supplement to known-motif analysis. Candidate footprints from the Buenrostro B-cell versus T-cell replicate comparison were exported to FASTA and analyzed with STREME in both contrast directions. The de novo-only branch yields candidate-derived motif families, with representative footprint-like aggregate profiles selected by negative center-versus-flank signal. The supplement branch appends the discovered motifs to a known database before diff-footprints; in the restricted-database sensitivity test, adding 16 candidate-derived motifs increased the highlighted motif-family count from 97 to 101, including 6 de novo-derived highlighted motifs. Tomtom labels indicate motif similarity to database entries. Thin aggregate lines show individual biological replicates and thick lines show condition means.

3.5. Replicate-aware Normalization

Replicate-aware differential footprint analysis adds an interpretable layer to condition comparison: repeated condition names define biological replicate groups, condition means are reported alongside individual replicate profiles, and optional sample-level quantile

normalization applies the same distribution-matching step before replicate group averages are compared. To show the effect on real data, we used the B-cell versus T-cell 2-vs-2 Buenrostro comparison and plotted aggregate corrected cut-site profiles for the same four motifs used in the biological consistency panel. The no-normalization and sample-quantile panels were extracted from the corresponding embedded diff-footprints HTML payloads, so the figure reflects the exact data used in the differential report. Thin lines show individual biological replicates and thick lines show condition means (Figure 5).

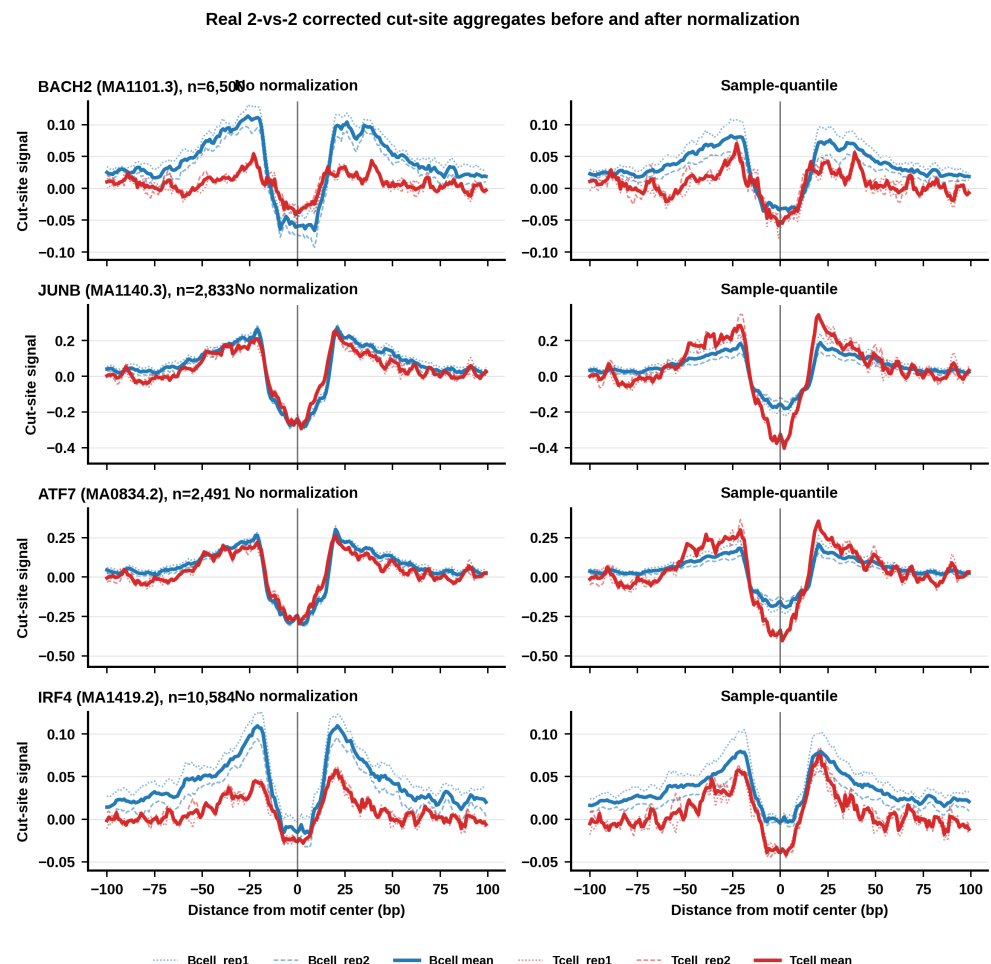


Figure 5. Effect of sample-level quantile normalization on real 2-vs-2 corrected cut-site aggregate profiles. BACH2, JUNB, ATF7, and IRF4 aggregates were extracted from the no-normalization and sample-quantile diff-footprints HTML reports for the Buenrostro B-cell versus T-cell replicate comparison. Thin lines show individual biological replicates, thick lines show condition means, and a lower center relative to the local flanks indicates stronger TF protection.

As a biological consistency check for the same 2-vs-2 Buenrostro replicate comparison, we inspected aggregate corrected cut-site profiles for selected motifs with visible footprint-like center depletion. BACH2 showed a B/GM12878-biased pattern in this dataset while remaining biologically non-exclusive because it is also reused in T-cell states. Conversely, JUNB and ATF7 showed stronger T-cell aggregate depletion; JUNB is an AP-1 family regulator of T-cell differentiation [11], and ATF7 is included here as a strong T-cell-deeper aggregate example rather than a lineage-specific marker. We also include IRF4 because TCR signaling can drive graded IRF4 expression in T cells [12], although its aggregate depletion is modest in this dataset. We treat these panels as an illustrative sanity check rather than

a marker-classification benchmark; a more negative center-versus-flank profile indicates stronger footprint-like protection at motif centers (Figure 6).

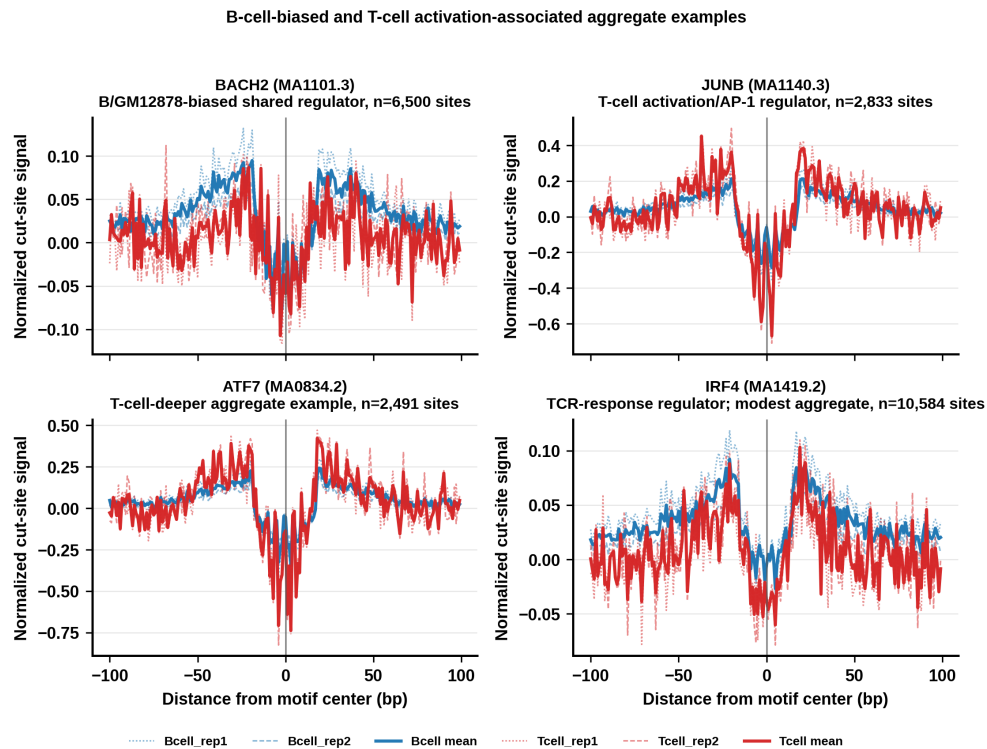


Figure 6. B-cell-biased and T-cell activation-associated aggregate examples from the B-cell versus T-cell replicate comparison. Aggregate profiles were generated from the same sample-quantile-normalized corrected cut-site payload embedded in the `diff-footprints` HTML report. Thin lines show individual biological replicates and thick lines show condition means. BACH2 provides the B/GM12878-biased shared-regulator example; JUNB and ATF7 show stronger T-cell aggregate depletion; IRF4 is included as a biologically motivated TCR-response regulator with a modest aggregate effect in this dataset.

3.6. Single-cell ATAC Fragments Produce Interpretable Pseudobulk Footprints

We evaluated pseudobulk aggregation using the public 10x Genomics PBMC granulocyte-sorted Multiome dataset [10]. The workflow used the official 10x ATAC fragments and GEX graph clusters, mapped 11,898 cells into broad immune pseudobulks, and retained seven groups after requiring at least 300 cells and 50,000 fragments: B cells, CD4 T cells, NK/cytotoxic T cells, CD14 monocytes, FCGR3A monocytes, dendritic cells, and a mixed myeloid group. The prepared figure inputs used chromosomes chr1–chr22 plus chrX. The kept pseudobulks contained 13.6–144.0 million counted fragments per group and were written as indexed fragment files plus CPM-normalized cut-site bigWigs. These outputs were readable by the same aggregate-plotting code used for bulk ATAC data.

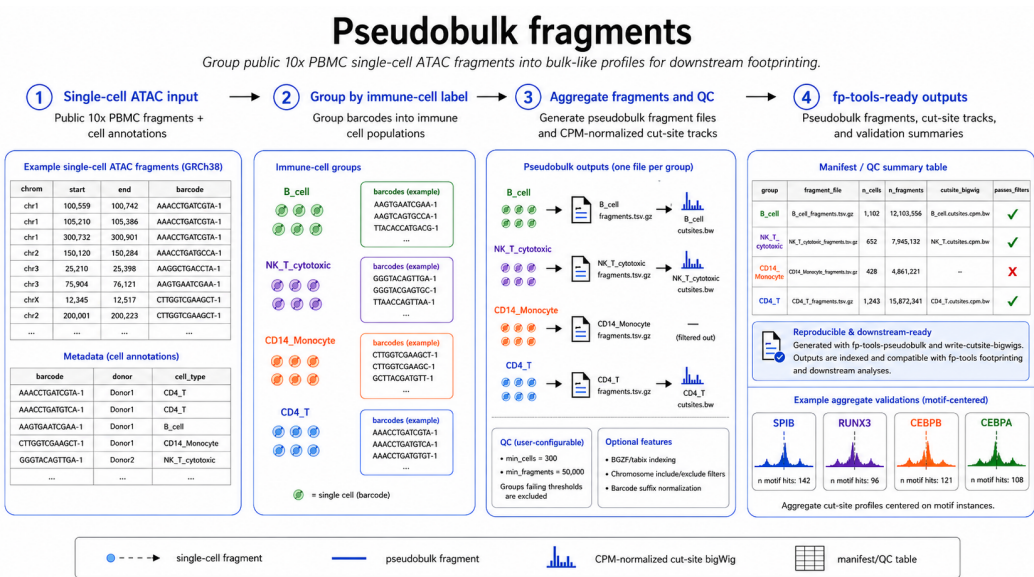


Figure 7. Example pseudobulk fragment workflow using public 10x PBMC single-cell ATAC data. Single-cell fragment records and cell annotations are grouped by broad immune-cell labels, converted into one pseudobulk fragment file per retained group, and summarized with QC fields, indexed outputs, and CPM-normalized cut-site bigWigs for downstream footprinting. The mini aggregate panels in the workflow illustrate motif-centered validation profiles from the same PBMC example, not a separate generic benchmark.

As a qualitative validation, we aggregated pseudobulk cut-site signal around exact motif centers rather than around broad peak centers. JASPAR2026 vertebrate motifs were scanned in the 10x ATAC peak set, motif hits were grouped by TF family, and candidate immune regulators were screened using a predeclared B-cell, T/NK-cell, and myeloid marker list. To avoid artificial fixed site counts in the figure, all motif hits were counted and used for each displayed TF. We selected SPIB as the B-cell-associated example, RUNX3 as the T/NK-associated example, and CEBPB and CEBPA as myeloid-associated examples from the predeclared immune-regulator screen. The fragment data and pseudobulk cut-site tracks were not modified; only the motif-centered site set used for aggregation changed. The resulting profiles show that single-cell fragments can be collapsed into group-level tracks and inspected with the same motif-centered aggregate workflow used for bulk ATAC data (Figure 8). The pseudobulk module can also emit pseudo-paired BAMs for downstream bias correction with zero read shifting.

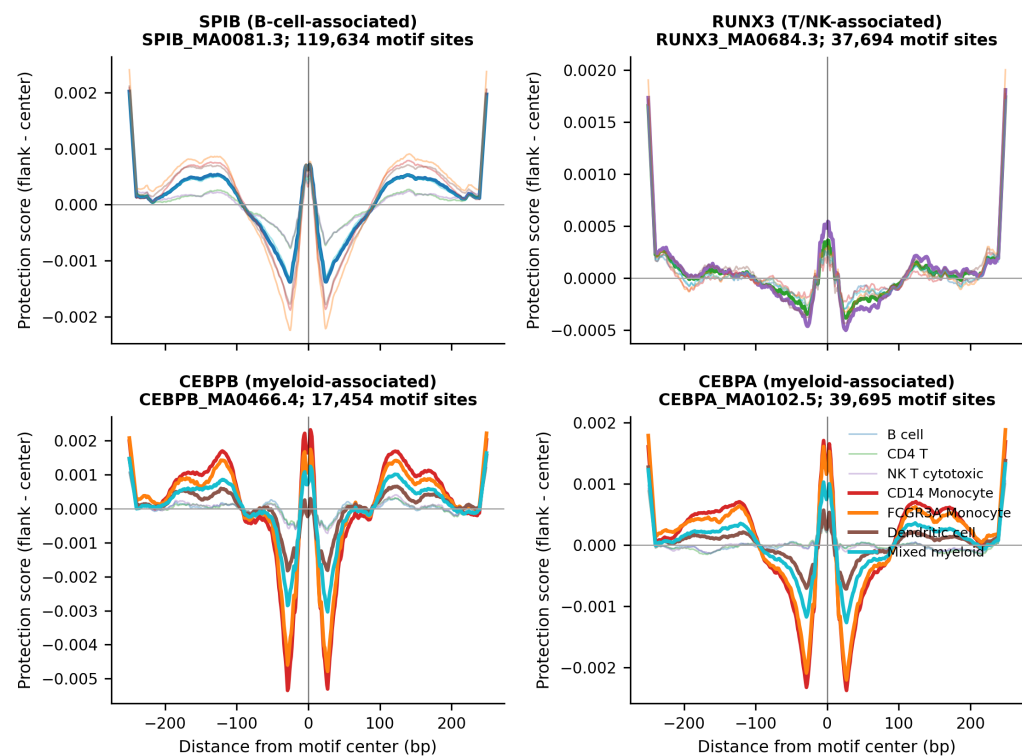


Figure 8. Motif-centered 10x PBMC pseudobulk protection-score aggregate profiles. Single-cell ATAC fragments from the public 10x PBMC Multiome dataset were grouped by broad immune-cell labels derived from official 10x gene-expression graph clusters, converted to CPM-normalized cut-site bigWigs, and aggregated around exact JASPAR2026 motif centers scanned inside 10x ATAC peaks. The plotted score is a flank-minus-center protection summary computed from the pseudobulk cut-site profile, so higher values indicate stronger local center depletion relative to the nearby flanks. SPIB, RUNX3, CEBPB, and CEBPA were selected by the predeclared lineage screen as B-cell-, T/NK-cell-, and myeloid-associated examples. Thin lines show all retained pseudobulk groups; thicker lines mark the expected pseudobulk groups for each lineage-associated TF. The source screen and natural motif-hit counts are reported in figures/supp_pseudobulk_tf_screen.tsv. The figure demonstrates that the pseudobulk fragment workflow produces group-level signal tracks compatible with motif-centered aggregate visualization.

3.7. Benchmark and Figure Reproducibility

The tiered benchmark design (Table 3), metric scripts, manifest builders, schema validator, engineering-runtime recorder, and figure-generation scripts provide an executable path from public inputs to publication-ready outputs. These resources serve three purposes. First, they verify that the classical footprinting workflow produces stable outputs on small fixtures and public-data subsets. Second, they provide matched-label inputs for bulk ATAC comparisons against TF ChIP-seq or CUT&RUN labels. Third, they support the manuscript demonstrations for de novo motif preparation, pseudobulk aggregation, replicate-aware normalization, and cut-site-resolution footprint calling.

Each benchmark run is expected to write a resolved manifest, software versions, parameters, and plotting tables next to the generated figures. This makes the figures auditable: a reader can inspect which samples, motifs, regions, and signal tracks were used, and the same tier can be rerun after code changes.

3.8. Footprinting Requires Cut-Site Resolution

Footprinting infers occupancy from a local depletion of Tn5 insertions at the bound site, flanked by hyper-accessible edges. We tested this directly on K562 CTCF (chromosome 4)

by computing a footprint-occupancy score at each peak’s best CTCF motif match—the mean signal in the flanks minus the mean signal at the motif site—using two different signal representations (Figure 9). On a smoothed fold-change-over-control coverage track the score is uninformative and in fact inverted (AUROC = 0.24), because coverage is *enriched*, not depleted, at bound accessible sites. When the same score is computed from base-resolution Tn5 cut sites derived from the ATAC BAM, it recovers a genuine footprint signal (AUROC = 0.68, 95% CI 0.65–0.70)—a large gain that arises purely from using the correct, cut-site-resolution signal that fp-tools’ bias-correction and footprint-scoring pipeline is built to produce. For CTCF, whose long motif is already highly predictive, the cut-site footprint is an orthogonal but weaker line of evidence and does not exceed the motif score; footprinting is expected to contribute most where motifs are weak or where condition-specific occupancy changes are the question, which the differential binding and replicate-aware modules target.

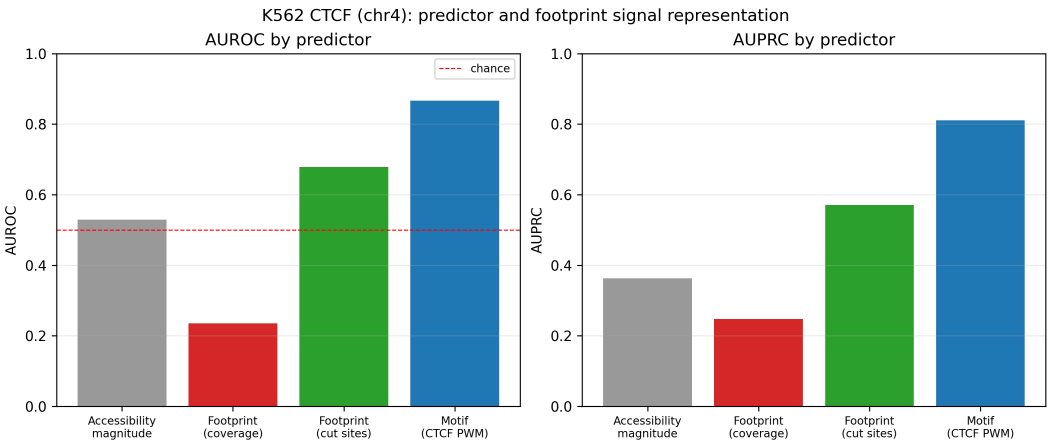


Figure 9. Predictor comparison for K562 CTCF (chromosome 4). A footprint-occupancy score is informative only at Tn5 cut-site resolution (green, AUROC 0.68); the same score on smoothed coverage is inverted and uninformative (red, 0.24). Raw accessibility (grey) is near chance and the CTCF motif (blue) is the single strongest predictor.

4. Discussion

fp-tools contributes a reproducible platform for ATAC-seq footprinting and motif-centered regulatory analysis. Its comparative strengths are interface consistency across bias correction, footprint calling, differential footprint analysis, and aggregate visualization; deterministic regression coverage; explicit handling of replicate-aware summaries; and an extension model in which de novo motif preparation and pseudobulk aggregation are integrated without disrupting the classical workflow (Table 4). These properties target the practical friction that slows applied ATAC-seq analysis and cross-method benchmarking today.

Table 4. Feature comparison across widely used ATAC-seq regulatory-analysis tools. ✓, native first-class support; ○, partial or indirect support; —, absent. `fp-tools` is an integrated, reproducible platform that combines classical footprinting with replicate-aware reporting, de novo motif-discovery preparation, and single-cell pseudobulk aggregation behind one command surface.

Feature	<code>fp-tools</code>	TOBIAS	HINT-ATAC	PRINT/ <code>scPrinter</code>	Chrom- BPNet	<code>maxATAC</code>
Bulk ATAC footprinting	✓	✓	✓	✓	○	—
Tn5 bias correction	✓	✓	✓	✓	✓	—
Classical footprint calling	✓	✓	✓	—	—	—
Motif-aware differential footprint analysis	✓	✓	○	○	—	—
De novo motif-discovery preparation	✓	—	—	✓	✓	—
<code>scATAC</code> / pseudobulk support	✓	○	○	✓	○	○
Replicate-aware reporting	✓	○	○	○	—	—
Visualization / reporting	✓	✓	○	✓	✓	○
YAML / batch execution	✓	—	—	—	—	—

Deep sequence-to-profile models such as ChromBPNet provide powerful sequence-driven occupancy and variant-effect analyses [6]. `fp-tools` addresses a complementary need: a lightweight, auditable classical footprinting workflow that can be run from corrected ATAC-seq signal tracks, inspected at motif-centered loci, and extended to replicate and pseudobulk settings. The replicate-aware summaries reported here use condition-level and replicate-level score variation to aid interpretation of differential footprint comparisons; they are designed for transparent reporting and visualization rather than as a replacement for specialized hierarchical models.

Limitations remain. The present benchmark evaluates discrimination within each cell-TF task under chromosome-held-out cross-validation; whole-genome cross-cell-line transfer (training on some cell lines and testing on others) and direct head-to-head runs of external tools such as TOBIAS, HINT-ATAC, and ChromBPNet on identical locus sets will require larger external-tool benchmark runs. Large public-benchmark inputs depend on external data acquisition and are therefore regenerated from committed manifests rather than stored directly in the repository. Natural next steps include larger external-tool comparisons, stronger per-replicate statistical models, and additional public validation datasets.

5. Conclusions

We presented `fp-tools`, a standalone, reproducible command-line platform that unifies classical ATAC-seq footprinting with motif-discovery preparation, pseudobulk aggregation, and replicate-aware reporting. By pairing stable, test-covered workflows with public-data manifests, source tables, and reproducible figure generation, the package lowers the barrier to standardized, benchmarkable regulatory-genomics analysis. Future work will add larger whole-genome transfer studies, direct external-tool comparisons, multiscale and nucleosome-aware validation, and stronger hierarchical differential-binding validation.

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Informed Consent Statement: Not applicable.

Data Availability Statement: fp-tools is openly available at <https://github.com/oncologylab/fp-tools>. All benchmark manifests, schemas, analysis scripts, and figure generators are included in the repository. Public input datasets are obtained from the ENCODE Project [13], the 10x Genomics PBMC Multiome example [10], and motif databases (e.g., JASPAR [14]) via the included discovery and download scripts. Manifests, scripts, small result tables, source tables, and figures are stored in the main repository. Reviewer reproduction instructions are provided in docs/reproduce-paper.md, with Conda and Docker execution files in environment.yml and the repository Dockerfile. Large reusable example-data files are distributed separately through the public oncologylab/fp-tools-data repository release assets, and full raw public inputs are regenerated from committed manifests rather than stored in version control.

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